

Exercise Sheet 5

Interior Point Methods and the Central Path

Algorithms and Geometry of Linear Programming — DIMAG Summer School, August 2026

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This sheet restates the four exercises of Lecture 5. Throughout, we work with the strictly feasible standard-form pair

$$\min c^\top x \text{ s.t. } \mathbf{A}x = b, x \geq 0, \quad \max b^\top y \text{ s.t. } \mathbf{A}^\top y + s = c, s \geq 0,$$

$\mathbf{A} \in \mathbb{R}^{m \times n}$ of full row rank. Products and quotients of vectors are componentwise. The *central path* $z(\mu) = (x(\mu), s(\mu), y(\mu))$, $\mu > 0$, is the unique feasible solution of the centrality equation $xs = \mu \mathbf{1}$, and $\mu(z) := x^\top s / n$ is the normalized duality gap. Newton steps solve

$$s \Delta x + x \Delta s = t, \quad \mathbf{A} \Delta x = 0, \quad \mathbf{A}^\top \Delta y + \Delta s = 0,$$

with target $t = \mu(z) \mathbf{1} - xs$ (*corrector*) or $t = -xs$ (*predictor*, also called the affine scaling direction, denoted $\Delta x^p, \Delta s^p$); the step moves to $(x + \alpha \Delta x, s + \alpha \Delta s, y + \alpha \Delta y)$ for a step size $\alpha \in (0, 1]$. As shown in the lecture, the scaled equations exhibit the step as an orthogonal decomposition: with $D := \text{diag}(\sqrt{x/s})$,

$$\underbrace{\sqrt{s/x} \Delta x}_{\in \ker(\mathbf{A}D)} + \underbrace{\sqrt{x/s} \Delta s}_{\in \text{im}((\mathbf{A}D)^\top)} = \frac{t}{\sqrt{xs}},$$

the two summands being the orthogonal projections of $t/\sqrt{xs}$ onto complementary subspaces. Exercise 1 concerns the *homogeneous self-dual embedding* of Ye, Todd and Mizuno: with the residual vectors of the all-ones point,

$$\bar{b} := b - \mathbf{A} \mathbf{1}, \quad \bar{c} := c - \mathbf{1}, \quad \bar{g} := c^\top \mathbf{1} + 1,$$

it is the LP

$$\begin{aligned} \min \quad & (n+1)\theta \\ \text{s.t.} \quad & \mathbf{A}x - b\tau + \bar{b}\theta = \mathbf{0} \\ & -\mathbf{A}^\top y + c\tau - \bar{c}\theta = s, \quad s \geq 0 \\ & b^\top y - c^\top x + \bar{g}\theta = \kappa, \quad \kappa \geq 0 \\ & -\bar{b}^\top y + \bar{c}^\top x - \bar{g}\tau = -(n+1), \end{aligned}$$

with $x \geq 0, \tau \geq 0$ and y, θ free; κ is a scalar variable (not the circuit imbalance), and the coefficient matrix acting on (y, x, τ, θ) is skew-symmetric. As shown in the lecture, following the central path of the embedding drives the objective $(n+1)\theta$ to 0, and its limit is an optimal solution with $\theta = 0$. Exercise 4 uses the straight line complexity of Lecture 4: $\text{SLC}_\gamma(x_i^m, g_0)$ is witnessed by a piecewise linear h with $\gamma x_i^m \leq h \leq x_i^m$ on $[0, g_0]$, and x_i^m is nonnegative, nondecreasing and concave.

Exercise 1 (The self-dual embedding). Verify the claims made in the lecture about the embedding:

- (a) check that $(y, x, \tau, \theta) = (\mathbf{0}, \mathbf{1}, 1, 1)$ is feasible, with slacks $(s, \kappa) = (\mathbf{1}, 1)$, and derive the gap identity $x^\top s + \tau \kappa = (n+1)\theta$ by multiplying the four constraint groups by y, x, τ, θ respectively and adding — all data terms cancel in skew-symmetric pairs;

- (b) show that if an optimal solution has $\tau > 0$, then $(x/\tau, s/\tau, y/\tau)$ is an optimal primal-dual pair for the original LP;
- (c) show that if instead $\kappa > 0$, then $\tau = 0$ and either $c^T x < 0$, in which case x certifies dual infeasibility via $Ax = 0, x \geq 0, c^T x < 0$, or $b^T y > 0$, in which case y certifies primal infeasibility via $A^T y \leq 0, b^T y > 0$.

Exercise 2 (Path rescaling). Let $z(1) = (x(1), s(1), y(1))$ be the central path point at $\mu = 1$. Show that rescaling the variables by $x_j \leftarrow x_j/x_j(1)$ (and correspondingly $A_j \leftarrow x_j(1) A_j, c_j \leftarrow x_j(1) c_j, s_j \leftarrow s_j/s_j(1)$) maps the instance to an equivalent one whose central path point at $\mu = 1$ is $(x, s) = (1, 1)$, and maps central path to central path.

Exercise 3 (Exact termination requires polarization). Let $z = (x, s, y)$ be strictly feasible and let $(\Delta x^P, \Delta s^P, \Delta y^P)$ be the predictor (affine scaling) direction at z .

- (a) Show that a predictor step with step size $\alpha = 1$ lands on a feasible optimal pair if and only if $\Delta x^P \Delta s^P = 0$ coordinatewise — i.e., the affine scaling direction is already *perfectly polarized*: on each coordinate, either $\Delta x_i^P = -x_i$ and $\Delta s_i^P = 0$, or the reverse. (Hint: Compute the products at $\alpha = 1$.)
- (b) Verify that the perfectly polarized pattern is consistent with the coordinatewise equation $\Delta x_i^P/x_i + \Delta s_i^P/s_i = -1$, and explain why the subspace constraints $\Delta x^P \in \ker(A), \Delta s^P \in \text{im}(A^T)$ — which force the step to be the *orthogonal* decomposition recalled above — generically rule it out. (The trust-region step of the lecture recovers the polarized pattern by imposing it through the constraint sets instead.)

Exercise 4 (Polarized segments from an SLC decomposition). Recall from Lecture 4 that the straight line complexity $\text{SLC}_\gamma(x_i^m, g_0)$ is witnessed by a piecewise linear h with $\gamma x_i^m \leq h \leq x_i^m$. Prove that each linear piece of a minimal h satisfies $h(g) = ag + b$ where $a, b \geq 0$. Conclude that up to a factor 2, each linear piece is either a *constant* or a *linear scaling* of the gap. Finally show that an SLC decomposition of all coordinates induces a partition of the path into $O(\sum_i \text{SLC}_\gamma(x_i^m))$ segments on which every coordinate either stays put or scales — the polarized segments of this lecture.